CS102 – Computer Programming I

Lab 1: Getting Started – IDE, First Program, Git & Number Systems

New Uzbekistan University (NewUU) | Summer 2026

Student Name: _____

Student ID: _____

Group / Section: _____

Learning goals

By the end of this lab, you should be able to:

- Install and configure **CLion** with a working C++ toolchain.
- Create, build, and run your **first C++ program**.
- Use `cout` and `cin` for simple output and input.
- Create a **Git** repository and push your code to GitHub.
- Convert numbers between **binary, decimal, octal and hexadecimal**.

Part A — Setting Up Your Environment

Task 1 Install CLion

Follow these steps:

a) Download and install CLion from `jetbrains.com/clion`.
b) Activate the free **student license**: visit `jetbrains.com/academy/student-pack`, register with your university e-mail, confirm, and log in inside CLion.
c) Make sure a C++ toolchain is detected (Windows: MinGW/MSVC; macOS: Xcode command-line tools; Linux: GCC/Clang).

Task 2 Create your first project

a) Open CLion → **New Project** → **C++ Executable**.
b) Name it `Lab1_YOUR_STUDENT_ID`.
c) CLion creates `main.cpp` (your program) and `CMakeLists.txt` (build configuration). Run the default program with the green **Run** button.

Part B — First Programs

Task 3 Hello, World!

Type, build, and run the classic first program:

```
#include <iostream>
using namespace std;

int main() {
    cout << "Hello, World!" << endl;
    return 0;
}
```

Then change the message to greet your group (e.g. `"Hello, CS102!"`) and run it again.

Task 4 Personal information

Write a program that asks the user for their **name** and **age**, then prints them back:

Example: Input: `Aziza 19` → Output: `Name: Aziza Age: 19`

Task 5 Three numbers on one line

Ask the user for three integers and print them on one line separated by spaces.

Example: Input: `4 7 9` → Output: `You entered: 4 7 9`

Task 6 Program banner

Print the following banner exactly (mind the line breaks):

```
========================
    CS102 PROGRAM
========================
```

Task 7 Student information card

Ask for **full name, student ID, major, GPA** and print:

```
-----------------------------------
     STUDENT INFORMATION
-----------------------------------
Name : Alisher Karimov
Student ID : 20260101
Major : Computer Science
GPA : 3.67
-----------------------------------
```

Hint: read the full name with `getline(cin, name)`.

Part C — Number Systems (by hand)

Task 8 Conversions

Convert each number to the requested base. Show your work.

1. $10110_2 = ?_{10}$
2. $110011_2 = ?_{10}$
3. $37_{10} = ?_2$
4. $58_{10} = ?_2$
5. $47_8 = ?_2$
6. $10110100_2 = ?_8$
7. $2C_{16} = ?_{10}$
8. $150_{10} = ?_{16}$

Task 9 Binary arithmetic

Compute in binary (show the carries/borrows):

1. $1101_2 + 1011_2$
2. $10110_2 + 1101_2$
3. $1100_2 - 101_2$
4. $10010_2 - 1011_2$

Part D — Git

Task 10 Your first repository

a) Create a GitHub account at `github.com` (use your university e-mail).
b) In CLion: **VCS** → **Enable Version Control Integration** → **Git**.
c) **Commit** all files with the message `First commit`.
d) Create a new **public** repository on GitHub and **Push** your project to it.
e) Useful terminal equivalents: `git init`, `git add .`, `git commit -m "msg"`, `git status`, `git push`.

Challenge

Modify the student-card program so that the GPA is printed with exactly two digits after the decimal point. *Hint:* look up `fixed` and `setprecision` from `<iomanip>`.